

LESTON REGO

Mangalore, Karnataka

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EDUCATION

Manipal School of Information Sciences, Manipal
M.E in Artificial Intelligence and Machine Learning

2025 – 2027

Sahyadri College of Engineering and Management, Mangalore
B.E in Computer Science and Engineering

2025

TECHNICAL SKILLS

Languages: Python, SQL

Machine Learning & Deep Learning: Regression, Classification, Predictive Modeling, Time-Series Forecasting.

Generative AI: Transformers, Attention Mechanisms, RAG (Retrieval-Augmented Generation).

Libraries & Frameworks: Scikit-learn, PyTorch, Pandas, NumPy, BeautifulSoup

Version Control: Git, GitHub

Data Visualization: Power BI, Matplotlib, Plotly

PROJECTS

YouTube Transcript Q&A Assistant (RAG) | Python, FAISS, SentenceTransformers, Groq LLM

- Built a RAG-based question-answering system that answers user queries from YouTube video transcripts using semantic search and LLM-generated responses.
- Extracted and cleaned YouTube subtitles using yt-dlp by removing duplicates, unnecessary tags, and formatting issues before processing the text.
- Converted transcript text into embeddings using Sentence Transformers and used FAISS for fast similarity-based retrieval of relevant transcript chunks.
- Implemented a two-stage retrieval pipeline using dense retrieval and CrossEncoder reranking to improve the relevance of retrieved results.
- Designed prompt workflows that generated answers only from transcript content to reduce hallucinations and improve factual accuracy.
- Developed an interactive Streamlit application for real-time question answering and viewing retrieved transcript chunks.

Inflation Prediction using ARIMA and NeuralProphet | Python, Time Series Forecasting

- Collected and preprocessed historical inflation data for time-series forecasting.
- Performed trend analysis and seasonality detection to understand inflation patterns over time.
- Built and trained ARIMA and NeuralProphet models to forecast future inflation trends.
- Compared predictions from statistical and deep learning models to analyze forecasting performance.
- Evaluated model accuracy by comparing predicted inflation values with actual data and visualized forecasting.

ECG Signal Reconstruction using Least Squares Optimization | Python, NumPy, Linear Algebra

- Developed an ECG signal reconstruction system using least squares optimization to recover missing signal values.
- Applied second-difference regularization techniques to maintain smooth and continuous ECG waveforms.
- Implemented matrix-based computations in NumPy for signal reconstruction and optimization.
- Compared original and reconstructed ECG signals using visualization techniques to evaluate reconstruction.